

如何使用

在 code 中 build 文件下使用 make 命令编译，编译成功后，在 build 文件夹下会生成可执行文件 P, 在 test 目录下的 test.josn 调整初始条件即可使用不同的测试用例。

2 维情形下运行较慢，因为在插值时对向量进行了多次复制，导致效率较低。

Problem $\exp(y + \sin(x))$

1 维 VCycle

规则边界，Dirichlet 边界条件，fullWeighting 松弛，linear 插值。

1. 网格数为 32。在第 6 次迭代时残差达到 10^{-6}

Residual norm at level 32: 0.000262642

Residual norm at level 16: 4.34873e-05

Residual norm at level 8: 8.95084e-06

6 iteration, error = 1.01905

2. 网格数为 64。在第 6 次迭代时残差达到 10^{-5}

Residual norm at level 64: 0.00178412

Residual norm at level 32: 0.000330479

Residual norm at level 16: 5.33846e-05

Residual norm at level 8: 1.07992e-05

6 iteration, error = 0.72165

3. 网格数为 128。在第 6 次迭代时残差达到 10^{-5}

Residual norm at level 128: 0.011126

Residual norm at level 64: 0.00208497

Residual norm at level 32: 0.000374178

Residual norm at level 16: 5.89672e-05

Residual norm at level 8: 1.16673e-05

6 iteration, error = 0.51057

4. 网格数为 256。在第 6 次迭代时残差达到 10^{-5}

Residual norm at level 256: 0.0661172

Residual norm at level 128: 0.0125497

Residual norm at level 64: 0.00228404

Residual norm at level 32: 0.000398829

Residual norm at level 16: 6.16598e-05

Residual norm at level 8: 1.20221e-05

6 iteration, error = 0.361112

规则边界, Dirichlet 边界条件, fullWeighting 松弛, quadratic 插值。

网格数为 256。在第 6 次迭代时残差达到 10^{-5}

Residual norm at level 256: 0.0488913

Residual norm at level 128: 0.0105387

Residual norm at level 64: 0.00206747

Residual norm at level 32: 0.000373161

Residual norm at level 16: 5.80517e-05

Residual norm at level 8: 1.01117e-05

6 iteration, error = 0.361111

说明二次插值的精度更高。

规则边界, Dirichlet 边界条件, injection 松弛, linear 插值。

1. 网格数为 128。在第 6 次迭代时残差达到 10^{-5}

Residual norm at level 128: 0.0500838

Residual norm at level 64: 0.01032

Residual norm at level 32: 0.00191325

Residual norm at level 16: 0.000303276

Residual norm at level 8: 5.87188e-05

6 iteration, error = 0.510574

2. 网格数为 256。在第 6 次迭代时残差达到 10^{-5}

Residual norm at level 256: 0.285242

Residual norm at level 128: 0.0599207

Residual norm at level 64: 0.0113986

Residual norm at level 32: 0.00202851

Residual norm at level 16: 0.000314789

Residual norm at level 8: 6.00298e-05

6 iteration, error = 0.361119

说明 fullweighting 松弛的精度更高。

1 维 FMGCycle

规则边界, Dirichlet 边界条件, fullWeighting 松弛, linear 插值。

1. 网格数为 32。在第 6 次迭代时残差达到 10^{-19}

Residual norm at level 8: 3.24181e-17

Residual norm at level 16: 1.69267e-16

Residual norm at level 8: 5.15177e-18

Residual norm at level 32: 3.10965e-16

Residual norm at level 16: 2.22317e-18

Residual norm at level 8: 1.40664e-19

6 iteration

2. 网格数为 64。在第 6 次迭代时残差达到 10^{-20}

Residual norm at level 8: 4.51756e-18
Residual norm at level 16: 1.04015e-16
Residual norm at level 8: 5.50852e-18
Residual norm at level 32: 1.46254e-16
Residual norm at level 16: 8.48956e-18
Residual norm at level 8: 3.76318e-19
Residual norm at level 32: 1.31426e-17
Residual norm at level 16: 1.40193e-19
Residual norm at level 8: 1.19572e-20
6 iteration

3. 网格数为 128。在第 6 次迭代时残差达到 10^{-21}

Residual norm at level 8: 3.78515e-18
Residual norm at level 16: 8.83714e-18
Residual norm at level 8: 2.54908e-19
Residual norm at level 32: 2.17648e-16
Residual norm at level 16: 1.77389e-17
Residual norm at level 8: 7.43627e-19
Residual norm at level 64: 2.92295e-16
Residual norm at level 32: 1.7813e-17
Residual norm at level 16: 1.23719e-18
Residual norm at level 8: 6.79429e-20
Residual norm at level 128: 9.51196e-16
Residual norm at level 64: 2.75592e-17
Residual norm at level 32: 3.43742e-19
Residual norm at level 16: 4.63851e-20
Residual norm at level 8: 3.62819e-21
6 iteration

4. 网格数为 256。在第六次迭代时残差达到 10^{-22}

Residual norm at level 8: 3.07089e-19
Residual norm at level 16: 1.14245e-17
Residual norm at level 8: 3.97293e-19
Residual norm at level 32: 1.91865e-17
Residual norm at level 16: 7.81932e-19
Residual norm at level 8: 2.82622e-20

Residual norm at level 64: 4.45551e-16
Residual norm at level 32: 3.70202e-17
Residual norm at level 16: 2.26092e-18
Residual norm at level 8: 9.54469e-20
Residual norm at level 128: 5.95322e-16
Residual norm at level 64: 3.63871e-17
Residual norm at level 32: 2.56007e-18
Residual norm at level 16: 2.04044e-19
Residual norm at level 8: 1.13026e-20
Residual norm at level 256: 1.94948e-15
Residual norm at level 128: 5.71161e-17
Residual norm at level 64: 7.40136e-19
Residual norm at level 32: 1.00687e-19
Residual norm at level 16: 1.08841e-20
Residual norm at level 8: 7.60319e-22
6 iteration

剩下的松弛方法和插值方法不再一一展示，运行结果均正确。

2 维 VCycle

规则边界，Dirichlet 边界条件，fullWeighting 松弛，linear 插值。

1. 网格数为 32。在第 11 次迭代时残差达到 10^{-1}

Residual norm at level 32: 5.74025
Residual norm at level 16: 1.64604
Residual norm at level 8: 0.262687
11 iteration, error = 28.2176

2. 网格数为 64。Residual norm at level 64: 62.6875

Residual norm at level 32: 19.2807
Residual norm at level 16: 4.80265
Residual norm at level 8: 0.694238
11 iteration, error = 56.6083

3. 网格数为 128。Residual norm at level 128: 446.068

Residual norm at level 64: 141.43
Residual norm at level 32: 37.8171
Residual norm at level 16: 8.28785
Residual norm at level 8: 1.10734
11 iteration, error = 113.376

2 维 FMGCycle

规则边界, Dirichlet 边界条件, fullWeighting 松弛, linear 插值。

1. 网格数为 32。在第 11 次迭代时残差达到 10^{-3}

Residual norm at level 8: 0.0048538

Residual norm at level 16: 0.0473602

Residual norm at level 8: 0.00300494

Residual norm at level 32: 0.194175

Residual norm at level 16: 0.0347955

Residual norm at level 8: 0.00288642

11 iteration, error = 32.054

2. 网格数为 64。在第 11 次迭代时残差达到 10^{-3}

Residual norm at level 8: 0.010503

Residual norm at level 16: 0.109764

Residual norm at level 8: 0.00677214

Residual norm at level 32: 0.450197

Residual norm at level 16: 0.0830837

Residual norm at level 8: 0.00642242

Residual norm at level 64: 1.83493

Residual norm at level 32: 0.34701

Residual norm at level 16: 0.0788803

Residual norm at level 8: 0.00615708

11 iteration, error = 64.0267

3. 网格数为 128。在第 11 次迭代时残差达到 10^{-3}

Residual norm at level 8: 0.00988042

Residual norm at level 16: 0.135318

Residual norm at level 8: 0.00750912

Residual norm at level 32: 0.587086

Residual norm at level 16: 0.10339

Residual norm at level 8: 0.00694519

Residual norm at level 64: 2.38983

Residual norm at level 32: 0.466767

Residual norm at level 16: 0.0979731

Residual norm at level 8: 0.00635629

Residual norm at level 128: 10.1725

Residual norm at level 64: 1.7243

Residual norm at level 32: 0.44599

Residual norm at level 16: 0.0945225

Residual norm at level 8: 0.00566489

11 iteration, error = 128.013

误差分析

误差较大的原因可能是矩阵组装出现失误，暂时没有找到原因。